(notes) 4.2 Linear Models

2. My Cat, Poots

She cost $125

Every month we buy her food:

- $60 + $20
- $30

And cat litter:

- $40

How many years will it take to spend $1 million on her? $10,000? We've had her for 2 years, how much did we spend on her?

How much $ will we spend on her if she lives to 10 years old?

What is the independent variable (x)?

What is the dependent variable (y)?

What is the base cost?

What is the rate of change (include units!)

Linear function:

Amount we spent on her?
(hint: use x=24 months because 24 months=2 years)

How long will it take to spend $1 million on Poots?
(hint: solve for x, when f(x)=1,000,000)

My Husband, plays games w his friends

The xbox cost $500. He bought a game for $70. Every month he pays $10 for game pass

It is been about a year since he got the xbox? How much has he spent?

How much will he spend in 2 years?

How long will it take (in years) for him to spend $10,000?

Should he get another game?

What is the independent variable (x)?

What is the dependent variable (y)?

What is the base cost?

What is the rate of change (include units!)

Linear function:

Amount he spent on xbox?
(hint: use x=12 months because 1 year =12 months)

Amount he spent on xbox...over 2 years?
(hint: again, convert the years to months and then plug in for x)

How long (in years) will it take for him to spend $10,000?
(hint: solve for x, when f(x)=10,000, then convert the months to years for me 🙂)

Name:

My Linear Function Word Problem

Situation:

Independent variable (x)

Dependent variable (y)

Base value?

Rate of change (include units!)

Equation of Linear Function, f(x)

Predicting question: evaluate f(x) for a value x=#

Solve for x, given f(x)=#, back solve for the x-val